Linear Algebra (MT-1004)

Homework#13

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Question 1

a) Compute the dot product of the vectors $\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$.

b) Given two vectors $\mathbf{a}$ and $\mathbf{b}$, show that $\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\|\|\mathbf{b}\| \cos\theta$.

c) Let $\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Compute the angle between the vectors.

d) Prove that the dot product is distributive: $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$.

e) **Python Task:** Write a Python function using NumPy that takes two vectors as input and returns the dot product and angle between them.